40009 37

Haskell TicTacToe

Submitters

kw1425

16
—
20

Emarking

```
Final Tests: Summary for kw1425 of c1
PPT 21
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  Public Tests:
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    student-tests/tictactoe-test/tictactoe/gameOver:        3 / 3
    student-tests/tictactoe-test/tictactoe/parsePosition:   6 / 6
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Git Repo: git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskelltictactoe_kw1425.git
Commit ID: 67e05
Git Diff: https://gitlab.doc.ic.ac.uk/lab2526_autumn/haskelltictactoe_kw1425/-/compare/master..67e05?from_project_id=133230
```

```haskell
module TicTacToe (gameOver, parsePosition, tryMove) where

import Text.Read (readMaybe)
import Data.Containers.ListUtils (nubOrd)
import Data.Char (isNumber)

import Control.Applicative
import Control.Monad

import Board
import Cell
import Player
import Helpers
```

2.5/3

```haskell
type Position = (Int, Int)

gameOver :: Board -> Bool
gameOver b = checkLines allLines
    where
        checkLines :: [[Cell]] -> Bool
        checkLines ls = any hasWon (map nubOrd ls)

        hasWon :: [Cell] -> Bool
        hasWon [x] = not (isEmpty x)  -- won if singleton is not empty
        hasWon _ = False

        allLines = rows b ++ cols b ++ diags b
```

→ instead of using ++, use any to check each one

```haskell
--
-- Moves must be of the form "row col" where row and col are integers
-- separated by whitespace. Bounds checking happens in tryMove, not here.
--
```

3/4

```haskell
parsePosition :: String -> Maybe Position
parsePosition xs = case words xs of
                   [a, b] -> (,) <$> readMaybe a <*> readMaybe b
                   _      -> Nothing
```

→ do [s1, s2] ← pure (words s) can let you pattern match without needing the case statement

↳ very good

```haskell
tryMove :: Player -> Position -> Board -> Maybe Board
tryMove player (i, j) (Board n cells) = do
    guard (i>=0 && i<n && j>=0 && j<n)      -- check bounds
    guard (isEmpty $ cells !! boardPos)     -- check if empty
    return (Board n updatedCells)
    where
        boardPos = i * n + j
        updatedCells = replace boardPos (Taken player) cells
```

2/2

```haskell
module Cell where

import Player (Player)

-- You may wish to define your own Show instance
data Cell = Empty | Taken Player deriving (Eq, Ord)

instance Show Cell where
    show :: Cell -> String
    show Empty = "-"
    show (Taken player) = show player

-- | Returns True if the given cell is Empty
isEmpty :: Cell -> Bool
isEmpty Empty = True
isEmpty _     = False
```

2/2

```haskell
module Board where

import Cell (Cell(..))
import Player (Player(..))

import Data.List (transpose)

data Board = Board Int [Cell] deriving (Eq, Show)

-- | Returns an empty board with sides of the given length
emptyBoard :: Int -> Board
emptyBoard n = Board n (replicate (n^2) Empty)

-- | Returns the rows of a given board.
rows :: Board -> [[Cell]]
rows (Board n cs) = rows' cs
  where rows' [] = []
        rows' cs = r : rows' rs
          where (r, rs) = splitAt n cs

-- | Returns the columns of a given board.
cols :: Board -> [[Cell]]
cols = transpose . rows

-- | Returns the diagonals of a given board.
diags :: Board -> [[Cell]]
diags (Board n cs) = [ [cs !! (k * (n + 1)) | k <- [0 .. n - 1]]
                     , [cs !! (k * (n - 1)) | k <- [1 .. n]]
                     ]

-------------------------------------------------------------------
testBoard1, testBoard2, testBoard3 :: Board
testBoard1 = Board 4 [ Taken O, Taken X, Empty,   Taken O
                     , Taken O, Empty,   Taken X, Taken X
                     , Taken O, Empty,   Empty,   Taken X
                     , Taken O, Taken X, Empty,   Empty
                     ]
testBoard2 = Board 2 [ Taken X, Empty
                     , Empty,   Empty
                     ]
testBoard3 = Board 5 [ Taken O, Taken X, Empty,   Taken O, Taken X
                     , Taken O, Empty,   Taken X, Taken X, Empty
                     , Empty,   Empty,   Taken X, Taken O, Taken O
                     , Taken O, Taken X, Empty,   Empty,   Taken X
                     , Taken X, Empty,   Taken O, Empty,   Empty
                     ]
```

```haskell
module Main where

import TicTacToe
import Board
import Cell
import Player
import Helpers

import Text.Read (readMaybe)
import Data.List (intersperse)
import Control.Monad

import System.IO (hSetBuffering, stdout, stdin, BufferMode(NoBuffering))

prettyPrint :: Board -> IO ()
prettyPrint b = putStrLn $ unlines (map joinRows rowList)
  where rowList = rows b
        joinRows :: [Cell] -> String
        joinRows = (intersperse ' ') . unwords . (map show)
```

2/2

```haskell
-- The following reflect the suggested structure, but you can manage the game
-- in any way you see fit.

-- You might like to use one of the following signatures, as per the spec.
-- but this is up to you.
--
-- doParseAction :: (String -> Maybe a) -> IO a
-- doParseAction :: String -> (String -> Maybe a) -> IO a

-- | Repeatedly read a target board position and invoke tryMove until
-- the move is successful (Just ...).
takeTurn :: Board -> Player -> IO Board
takeTurn b p = do
  putStrLn ("Player " ++ show p)
  putStrLn "Enter the grid position to play ensuring there is a space between the row and column number: "
  pos <- getLine
  case parsePosition pos of
    Nothing ->
      do
      putStrLn "Enter the position in the correct format!"
      takeTurn b p
    Just position ->
      do
      case tryMove p position b of
        Nothing ->
          do
          putStrLn "Invalid move. Try again."
          takeTurn b p
        Just updatedBoard -> return updatedBoard
```

1.5/3

try extracting this into a helper function. Currently, you print the beginning lines repeatedly on an invalid input

↳ use your definition of doParseAction

```haskell
-- | Manage a game by repeatedly: 1. printing the current board, 2. using
-- takeTurn to return a modified board, 3. checking if the game is over,
-- printing the board and a suitable congratulatory message to the winner
-- if so.
playGame :: Board -> Player -> IO ()
playGame b p = do
  prettyPrint b
  newBoard <- takeTurn b p
  if gameOver newBoard then
    do
      putStrLn "Game Over!"
      putStrLn ("Player " ++ show p ++ " won!")
      prettyPrint newBoard
  else
    playGame newBoard (swap p)
```

2/2


```haskell

-- | Print a welcome message, read the board dimension, invoke playGame and
-- exit with a suitable message.
doParseAction :: (String -> Maybe a) -> IO a
doParseAction f = do
  line <- getLine
  case f line of
    Nothing -> do
      putStrLn "Invalid value. Please try again."
      doParseAction f
    Just val -> return val
```

1/2

```haskell
main :: IO ()
main = do
  disableBuffering -- don't remove!
  putStrLn "Welcome to Tic Tac Toe!"
  putStrLn "Enter the board size (NxN): "
  n <- doParseAction readMaybe
  playGame (emptyBoard n) startingPlayer
  return ()
```

you need to check the dimensions are valid (use filter.Maybe)

```haskell

{-|
When ran via 'cabal run', Haskell will "buffer" the input and output streams
for performance. This is annoying, since it means that some printing can happen
out-of-order -- the strings are only written when a newline is entered!

There are a few ways of getting around this, like using 'hFlush stdout' after
each print. However, in this case, it is simplest to just disable the buffering
altogether. This function *must* be called at the start of your 'main' function.
-}
disableBuffering :: IO ()
disableBuffering = do
  hSetBuffering stdout NoBuffering
  hSetBuffering stdin NoBuffering
```

```haskell
prompt :: String -> (String -> Maybe a) -> IO a
prompt msg validate = until Just M (validate <$> putStr msg *>

untilJustM :: Monad m => m (Maybe a) -> m a
untilJustM mmse = mmse >>= Maybe(untilJustM mmse) return
```

```
-------- Test Output --------
Config file path source is default config file.
Config file not found: /root/.config/cabal/config
Writing default configuration to /root/.config/cabal/config
Resolving dependencies...
Build profile: -w ghc-9.6.7 -O1
In order, the following will be built (use -v for more details):
 - call-stack-0.4.0 (lib) (requires build)
 - colour-2.3.6 (lib) (requires build)
 - prettyprinter-1.7.1 (lib) (requires build)
 - tagged-0.8.9 (lib) (requires build)
 - tictactoe-0.1.0.0 (lib) (first run)
 - ansi-terminal-types-1.1.3 (lib) (requires build)
 - tictactoe-0.1.0.0 (exe:game) (first run)
 - ansi-terminal-1.1.3 (lib) (requires build)
 - prettyprinter-ansi-terminal-1.1.3 (lib) (requires build)
 - optparse-applicative-0.19.0.0 (lib) (requires build)
 - tasty-1.5.3 (lib) (requires build)
 - tasty-hunit-0.10.2 (lib) (requires build)
 - tictactoe-0.1.0.0 (test:tictactoe-test) (first run)
Starting     tagged-0.8.9 (lib)
Starting     call-stack-0.4.0 (lib)
Starting     colour-2.3.6 (lib)
Starting     prettyprinter-1.7.1 (lib)
Building     call-stack-0.4.0 (lib)
Building     colour-2.3.6 (lib)
Building     prettyprinter-1.7.1 (lib)
Building     tagged-0.8.9 (lib)
Installing   call-stack-0.4.0 (lib)
Completed    call-stack-0.4.0 (lib)
Configuring library for tictactoe-0.1.0.0...
Installing   tagged-0.8.9 (lib)
Completed    tagged-0.8.9 (lib)
Preprocessing library for tictactoe-0.1.0.0...
Building library for tictactoe-0.1.0.0...
[1 of 5] Compiling Helpers          ( src/Helpers.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Helpers.o, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Helpers.dyn_o )
[2 of 5] Compiling Player           ( src/Player.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Player.o, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Player.dyn_o )
[3 of 5] Compiling Cell             ( src/Cell.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Cell.o, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Cell.dyn_o )
[4 of 5] Compiling Board            ( src/Board.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Board.o, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/Board.dyn_o )
Installing   colour-2.3.6 (lib)
[5 of 5] Compiling TicTacToe        ( src/TicTacToe.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/TicTacToe.o, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/build/TicTacToe.dyn_o )
Completed    colour-2.3.6 (lib)
Starting     ansi-terminal-types-1.1.3 (lib)
Configuring executable 'game' for tictactoe-0.1.0.0...
Building     ansi-terminal-types-1.1.3 (lib)
Preprocessing executable 'game' for tictactoe-0.1.0.0...
Building executable 'game' for tictactoe-0.1.0.0...
[1 of 1] Compiling Main             ( app/Main.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/x/game/build/game/game-tmp/Main.o )
[2 of 2] Linking /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/x/game/build/game/game
Installing   ansi-terminal-types-1.1.3 (lib)
Completed    ansi-terminal-types-1.1.3 (lib)
Starting     ansi-terminal-1.1.3 (lib)
Building     ansi-terminal-1.1.3 (lib)
Installing   ansi-terminal-1.1.3 (lib)
Completed    ansi-terminal-1.1.3 (lib)
Installing   prettyprinter-1.7.1 (lib)
```

```
Completed    prettyprinter-1.7.1 (lib)
Starting     prettyprinter-ansi-terminal-1.1.3 (lib)
Building     prettyprinter-ansi-terminal-1.1.3 (lib)
Installing   prettyprinter-ansi-terminal-1.1.3 (lib)
Completed    prettyprinter-ansi-terminal-1.1.3 (lib)
Starting     optparse-applicative-0.19.0.0 (lib)
Building     optparse-applicative-0.19.0.0 (lib)
Installing   optparse-applicative-0.19.0.0 (lib)
Completed    optparse-applicative-0.19.0.0 (lib)
Starting     tasty-1.5.3 (lib)
Building     tasty-1.5.3 (lib)
Installing   tasty-1.5.3 (lib)
Completed    tasty-1.5.3 (lib)
Starting     tasty-hunit-0.10.2 (lib)
Building     tasty-hunit-0.10.2 (lib)
Installing   tasty-hunit-0.10.2 (lib)
Completed    tasty-hunit-0.10.2 (lib)
Configuring test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Preprocessing test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Building test suite 'tictactoe-test' for tictactoe-0.1.0.0...
[1 of 2] Compiling IC.Exact         ( test/IC/Exact.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/t/tictactoe-test/build/tictactoe-test/tictactoe-test-tmp/IC/Exact.o )
[2 of 2] Compiling Main             ( test/Tests.hs, /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/t/tictactoe-test/build/tictactoe-test/tictactoe-test-tmp/Main.o )
[3 of 3] Linking /tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/t/tictactoe-test/build/tictactoe-test/tictactoe-test
Resolving dependencies...
Build profile: -w ghc-9.6.7 -O1
In order, the following will be built (use -v for more details):
 - tictactoe-0.1.0.0 (lib) (configuration changed)
 - tictactoe-0.1.0.0 (test:tictactoe-test) (configuration changed)
Configuring library for tictactoe-0.1.0.0...
Preprocessing library for tictactoe-0.1.0.0...
Building library for tictactoe-0.1.0.0...
Configuring test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Preprocessing test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Building test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Running 1 test suites...
Test suite tictactoe-test: RUNNING...
tictactoe
  emptyBoard
    #1: OK
    #2: OK
    #3: OK
  gameOver
    #1: OK
    #2: OK
    #3: OK
  parsePosition
    #1: OK
    #2: OK
    #3: OK
    #4: OK
    #5: OK
    #6: OK
  tryMove
    #1: OK
    #2: OK
    #3: OK
    #4: OK
    #5: OK

```

```
All 17 tests passed (0.00s)

Test suite tictactoe-test: PASS
Test suite logged to:
/tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/t/tictactoe-test/test/tictactoe-0.1.0.0-tictactoe-test.log
1 of 1 test suites (1 of 1 test cases) passed.
copying test from skeleton
Resolving dependencies...
Build profile: -w ghc-9.6.7 -O1
In order, the following will be built (use -v for more details):
 - tictactoe-0.1.0.0 (lib) (configuration changed)
 - tictactoe-0.1.0.0 (test:tictactoe-test) (configuration changed)
 - tictactoe-0.1.0.0 (exe:game) (dependency rebuilt)
Configuring library for tictactoe-0.1.0.0...
Preprocessing library for tictactoe-0.1.0.0...
Building library for tictactoe-0.1.0.0...
Configuring test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Preprocessing executable 'game' for tictactoe-0.1.0.0...
Building executable 'game' for tictactoe-0.1.0.0...
Preprocessing test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Building test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Resolving dependencies...
Build profile: -w ghc-9.6.7 -O1
In order, the following will be built (use -v for more details):
 - tictactoe-0.1.0.0 (lib) (configuration changed)
 - tictactoe-0.1.0.0 (test:tictactoe-test) (configuration changed)
Configuring library for tictactoe-0.1.0.0...
Preprocessing library for tictactoe-0.1.0.0...
Building library for tictactoe-0.1.0.0...
Configuring test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Preprocessing test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Building test suite 'tictactoe-test' for tictactoe-0.1.0.0...
Running 1 test suites...
Test suite tictactoe-test: RUNNING...
tictactoe
  emptyBoard
    #1: OK
    #2: OK
    #3: OK
  gameOver
    #1: OK
    #2: OK
    #3: OK
  parsePosition
    #1: OK
    #2: OK
    #3: OK
    #4: OK
    #5: OK
    #6: OK
  tryMove
    #1: OK
    #2: OK
    #3: OK
    #4: OK
    #5: OK

All 17 tests passed (0.00s)

Test suite tictactoe-test: PASS
Test suite logged to:
```

```
/tmp/d20251107-34-utnjx5/dist-newstyle/build/x86_64-linux/ghc-9.6.7/tictactoe-0.1.0.0/t/tictactoe-test/test/tictactoe-0.1.0.0-tictactoe-test.log
1 of 1 test suites (1 of 1 test cases) passed.

-------- Test Errors --------
Warning: The package list for 'hackage.haskell.org' is 44 days old.
Run 'cabal update' to get the latest list of available packages.
Warning: The package list for 'hackage.haskell.org' is 44 days old.
Run 'cabal update' to get the latest list of available packages.
Warning: The package list for 'hackage.haskell.org' is 44 days old.
Run 'cabal update' to get the latest list of available packages.
Warning: The package list for 'hackage.haskell.org' is 44 days old.
Run 'cabal update' to get the latest list of available packages.
```